Sample Question Paper

Mathematics

Class: X

Time allowed: 3 hours Maximum Marks: 80

Name: …… Roll No.: ……

General Instructions:

1. This question paper contains four sections: A, B, C and D.
2. All questions are compulsory, except where an internal choice is given.
3. Show your working in Sections B, C and D. Marks are awarded for steps.
4. Use pi = 22/7 wherever required, unless the question states otherwise.

Section A – Objective Questions (20 marks)

Choose the correct option for each question.

Q1. The HCF of 12 and 18 is: (1 mark)

(a) 3
(b) 6
(c) 9
(d) 12

Q2. The zero of the polynomial p(x) = x - 5 is: (1 mark)

(a) -5
(b) 0
(c) 5
(d) 1

Q3. The pair of linear equations 2x + 3y = 5 and 4x + 6y = 10 has: (1 mark)

(a) a unique solution
(b) no solution
(c) infinitely many solutions
(d) exactly two solutions

Q4. The roots of the equation x^2 - 3x - 10 = 0 are: (1 mark)

(a) 5 and 2
(b) -5 and 2
(c) 5 and -2
(d) -5 and -2

Q5. The 10th term of the AP 2, 5, 8, ... is: (1 mark)

(a) 26
(b) 29
(c) 32
(d) 35

Q6. The distance between the points (0, 0) and (6, 8) is: (1 mark)

(a) 8 units
(b) 9 units
(c) 10 units
(d) 14 units

Q7. The value of sin^2 A + cos^2 A is always: (1 mark)

(a) 0

(b) 1

(c) 2

(d) depends on the value of the angle

Q8. The value of tan 45 degrees is: (1 mark)

(a) 0

(b) 1

(c) square root of 3

(d) not defined

Q9. A fair coin is tossed once. The probability of getting a head is: (1 mark)

(a) 0

(b) 1/4

(c) 1/2

(d) 1

Q10. The mode of the data 2, 3, 3, 5, 7, 3, 9 is: (1 mark)

(a) 3

(b) 5

(c) 7

(d) 2

Q11. The area of a circle of radius 7 cm is: (1 mark)

(a) 44 cm^2

(b) 154 cm^2

(c) 308 cm^2

(d) 616 cm^2

Q12. The sum of the first 10 natural numbers is: (1 mark)

(a) 45

(b) 50

(c) 55

(d) 100

Q13. The discriminant of the equation x^2 + 4x + 4 = 0 is: (1 mark)

(a) 16

(b) 8

(c) 0

(d) -16

Q14. The common difference of the AP 7, 3, -1, ... is: (1 mark)

(a) 4

(b) -4

(c) 3

(d) -3

Q15. The areas of two similar triangles are in the ratio 4 : 9. The ratio of their corresponding sides is:(1 mark)

(a) 2 : 3

(b) 4 : 9

(c) 16 : 81

(d) 3 : 2

Q16. The value of cos 60 degrees is: (1 mark)

(a) square root of 3 upon 2

(b) 1/2

(c) 1

(d) 0

Q17. The mean of the first five even natural numbers is: (1 mark)

(a) 4

(b) 5

(c) 6

(d) 8

Q18. The number of tangents that can be drawn to a circle from a point outside it is: (1 mark)

(a) 0

(b) 1

(c) 2

(d) infinitely many

Q19. The volume of a cube of side 3 cm is: (1 mark)

(a) 9 cm^3

(b) 18 cm^3

(c) 27 cm^3

(d) 81 cm^3

Q20. The mid-point of the line segment joining (2, 4) and (6, 8) is: (1 mark)

(a) (3, 5)

(b) (4, 6)

(c) (8, 12)

(d) (4, 4)

Section B – Short Answers I (18 marks)

Answer each question in brief, showing your working.

Q21. Solve the following pair of linear equations by the substitution method: x + y = 10 and x - y = 4. (3 marks)

Q22. Find a quadratic polynomial whose zeroes are 3 and -2. (3 marks)

Q23. Find the sum of the first 20 terms of the AP 5, 8, 11, ... (3 marks)

Q24. Prove that 1 + tan^2 A = sec^2 A. (3 marks)

Q25. Find the area of a sector of a circle of radius 14 cm, where the angle of the sector is 90 degrees. (3 marks)

Q26. Find the median of the following data: 12, 15, 11, 20, 18, 14, 16. (3 marks)

Section C – Short Answers II (24 marks)

Answer each question, showing all the steps of your working.

Q27. Divide the polynomial 3x^2 + 5x - 4 by x + 2 and write down the quotient and the remainder. (4 marks)

Q28. Find the coordinates of the point which divides the line segment joining (2, -3) and (8, 3) internally in the ratio 2 : 1. (4 marks)

Q29. In triangle ABC, a line DE is drawn parallel to BC, with D on AB and E on AC. If AD = 3 cm, DB = 6 cm and AE = 4 cm, find EC. (4 marks)

Q30. From a point on the ground, 30 m away from the foot of a tower, the angle of elevation of the top of the tower is 60 degrees. Find the height of the tower. (4 marks)

Q31. A solid cylinder has radius 7 cm and height 20 cm. Find its total surface area. (4 marks)

Q32. Two dice are thrown at the same time. Find the probability that the sum of the numbers appearing on them is 8. (4 marks)

Section D – Long Answers (18 marks)

Answer each question in detail. Internal choice is given in Q34.

Q33. The area of a rectangular plot is 528 m^2. The length of the plot is one more than twice its breadth. Find the length and the breadth of the plot. (6 marks)

Q34. Prove that the lengths of the tangents drawn from an external point to a circle are equal. (6 marks)

OR

Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre of the circle. (6 marks)

Q35. The following distribution shows the daily pocket allowance (in rupees) of the children of a locality. Find the mean daily pocket allowance. Daily allowance: 11-13 (7 children), 13-15 (6), 15-17 (9), 17-19 (13), 19-21 (20), 21-23 (5), 23-25 (4). (6 marks)